

Enhanced Vulnerable Pedestrian Detection using Deep Learning

Zahid Ahmed, R. Iniyavan and Madhan Mohan P

Abstract—Road forms an integral part of quotidian commute and yet remains equally unsafe for commuters especially for pedestrians. Among the various categories that exists for object detection, detecting pedestrians remain a daunting task owing to a large changeability in the background as well as severe overlapping or occlusion. Also achieving high accuracy and real time speed pose a serious challenge since pedestrian detection should be both accurate as well as fast enough to be deployed in real time. This paper addresses these requirements through the usage of depth wise separable convolution and Single Shot Detector framework employing different activation maps using OpenCV to achieve a reliable, robust and competent deep learning based pedestrian detection for real time operations.

Index Terms—Deep Learning, MobileNets, Pedestrian Detection, Single Shot Detectors.

I. INTRODUCTION

DEEP learning is interdisciplinary by nature and allows to integrate the concepts of biology, computer science and neuromorphic systems. In order to understand how a deep neural network actually works it is very important to first dig deeper into the abyss of our very own neural networking system and decipher the complex series of events that happens when a human mind starts to identify objects, labels them effortlessly and learns to distinguish one from another.

While artificial intelligence reproduce the behavioral pattern of a human being, machine learning not just simply mimics the behavior but also imitates how humans learn which is a subtle difference worth noting. Deep learning on the other hand is a more sophisticated version of machine learning that has higher accuracy and more intensive computations that provides solutions to some of the most intricate areas of artificial intelligence.

Pedestrian detection is integral for comprehending pedestrian response and recognition with pedestrian profiling. One of the most influential pedestrian detection system comes from the work of Dalal[1] a proponent of histogram of gradient alongside support vector machine algorithm.

Dollar et al put forward the features of the channel[2] and the collected channel features[3] that made use of gradient of histogram as well as magnitude to improvise pedestrian features along with performance[4]. The application of MobileNets to pedestrian detection is explored to yield better results by Hyok Song and exceptional results on public datasets were achieved using SSD[5].

Fig. 1 depicts a typical Deep learning setup that provides an exhaustive picture about the sequence of steps that follows when a deep neural network is deployed to undergo training and validation to produce an output that best suits an application. Deep Learning setups are usually very robust since they are built from a strong interconnection of a vast number of hidden layers that serve to adjust the weights and bias associated with each input to predict the output by minimizing error while maximizing the probability of correct decision. The following are the major components that are defined for a typical deep learning setup

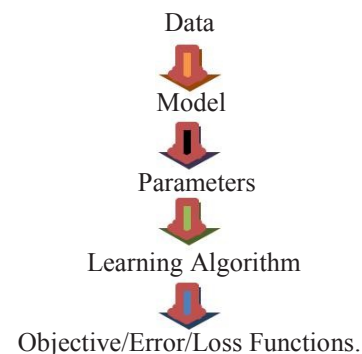


Fig. 1. A typical Deep Learning Setup

Any deep neural network largely fits into this network fairly easily and in most cases satisfies the objective function of employing a deep neural network given a specific application that one wishes to use the network to produce an output for.

A. Data

One of the most fundamental and integral part of any deep learning setup is the data that is fed to a deep neural network. The data can be of different types such as audio data, text data, image data or video data. However for a deep neural network to successfully be able to decipher the data and perform meaningful operation the data needs to be ideal in few aspects such as

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1.Relevancy - The data that is used in a deep neural network must be relevant to the application. Neural networks are innately naive and they learn what is taught to them. The training data should bear a resemblance to the real-world data to be able to classify and predict.

2.Proper Classification - An important attribute of a good data is well defined labels and annotations to classify what is what. Annotating the data especially when the data set is very huge can be a cumbersome procedure but yields outstanding results when performed with accuracy and precision.

3.Formatting - In neural networks data need to be converted into vectors and not just that but also vectors of the same length when they enter the neural net. Hence the data should necessarily be of same size. This is achieved during pre-processing.

4.Accessibility - An ideal data is one that can be easily accessed. If the data is placed over several unconnected databases then building data pipelines while pre processing would be extremely time consuming which would hinder the progress of building a deep neural network.

B. Model

The model is the approximation of the relationship that may exist amidst the input and output variables. The basic advantage of this component is the flexibility since new models can be derived and defined from pre existing pre trained and pre tested models or for the better of it even the development of new models is a sheer possibility. Arriving at an approximation of a relation is not the only task at hand. The approximation should establish an optimized and general relationship between the input and output variables. It has to be kept in mind that the value of a true function, which gives a relationship between these variables is never really known. This is true since the prior knowledge about the function would give an output whenever we plug an input to the known function which is not the case in deep learning.

C. Parameters

A model almost always contains parameters on which the approximated function is dependent. These parameters need to be learned and understood by the network so as to be able to successfully implement the network to produce meaningful results.

D. Learning Algorithms

The parameters in the model that are defined and assumed need to be learned by the network via a process or a set of rules for the problem solving operation. This is termed as the learning algorithm of the model.

E. Objective/Loss/Error functions

Objective functions help realize objective or the goal that is highly variable and dependent on task at hand. The SSD network model uses different activation maps to predict chases and hence is ideal for real-time detection of pedestrians. As such, the SSD network model will be utilized detection capabilities. The highlight of this paper is the detection accuracy with a faster FPS throughput[6-8].

II. DATA SELECTION AND NETWORK SETUP

A. Data

With the availability of large number of datasets it is often hard to decipher which dataset to choose from. In our case we choose MS-COCO owing to its large number of categorized and labeled data thus making recognition in context easy and smooth.

B. Network Architecture

Deep Learning most commonly employs three types of object detection techniques – the RCNN, YOLO or the SSD. RCNNs, as shown in Fig. 2, are accurate but they compromise speed. Furthermore the RCNN algorithms are relatively slower due to generation of region proposals prior to classification as depicted below.

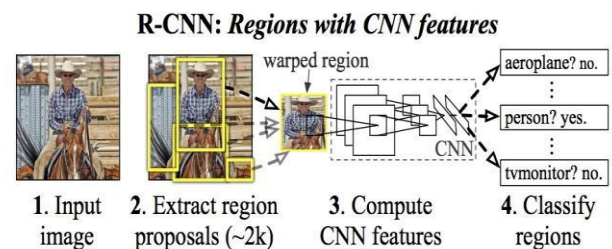


Fig. 2. Region proposal in RCNN

Also You Only Look Once (YOLO) although extremely fast in detection process are only far from being accurate which is a necessity in real time operations, any event of misclassification could jeopardize lives and hence not a feasible solution. YOLO framework is shown in Fig. 3.

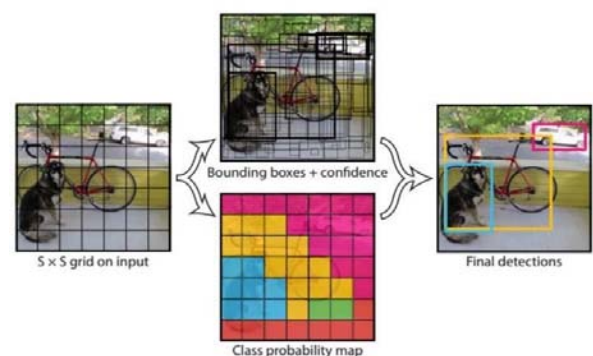


Fig. 3. YOLO framework

SSD frameworks depicted in Fig. 4 provide the best results with a balance of both speed and accuracy and without compromising any one to the extent of rendering the output meaningless in real time.

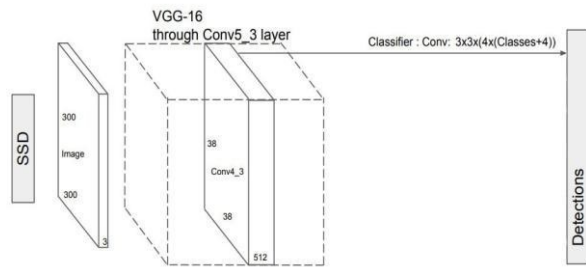


Fig. 4. Single Shot Detector architecture

SSDs can also help achieve a modest FPS suitable for any real time application and therefore it is an ideal choice.

C. Network Architecture

Network architecture need to be robust. In a majority of applications existing architectures are pre-defined such as ResNet or VGG but given the constraints, the large size of such network architecture is not well tolerated by devices with limited resources and hence MobileNets serve as ideal architectures in such devices.

MobileNets can be distinguished from conventional CNNs in that they use separable convolution. This is achieved by depth wise convolution followed by point wise convolution as shown in Fig. 5.

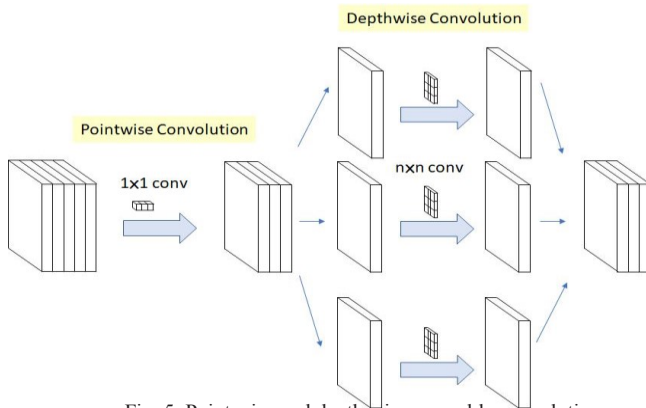


Fig. 5. Point wise and depth wise separable convolution

D. Software Description

Deep learning networks can be implemented in various software platforms. However, Anaconda being one of the most convenient platform with several built in IDEs is chosen. One major advantage of using this platform is the existence of built in libraries that are integrated and fully connected that runs with packages and libraries specific to our applications. Computational modelling is also much easier with Anaconda and hence a preferred choice[9,10].

III. EXPERIMENTAL RESULTS AND ANALYSIS

During the experiments, the software and the hardware that were utilized for realizing the objective is illustrated in Table I below.

TABLE I
PLATFORM DESCRIPTION

Hardware Platform		Software Platform	
CPU	AMD A8-6410 64-bit operating system x64based processor	Deep Learning Framework	SSD
		Network Architecture	MobileNet
		Operating System	Windows10

The Windows operating system on an AMD-A8 x64 based processor was used for setting up the platform i.e. Anaconda an open source platform that supports convenient and hassle free package deployment and management. Anaconda comes with built in Integrated Development Environment that is used for implementing OpenCV that drives computer vision for complex and intelligent machine recognition applications.

Name	Type	Size	Value
i	int32	1	0
idx	int	1	5
key	int	1	113
label	str	1	person: 99.99%
rval	bool	1	True
startX	int32	1	72
startY	int32	1	-2

Fig. 6. Target Detection scores

Fig. 6 illustrates that the model successfully detects the person class trained for pedestrian detection with a high degree of accuracy i.e 99.99% and FPS of 13-14

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Console 4/A
Python 3.6.8 |Anaconda, Inc.| (default, Feb 21 2019, 18
Type "copyright", "credits" or "license" for more infor
IPython 7.3.0 -- An enhanced Interactive Python.

In [1]: runfile('C:/Users/naresh-s/Desktop/wc.py', wdir

In [2]: runfile('C:/real-time-object-detection/real_tim
[INFO] loading model...
[INFO] starting video stream...
[INFO] elapsed time: 23.80
[INFO] approx. FPS: 13.49

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Fig. 7. Frames per second throughput

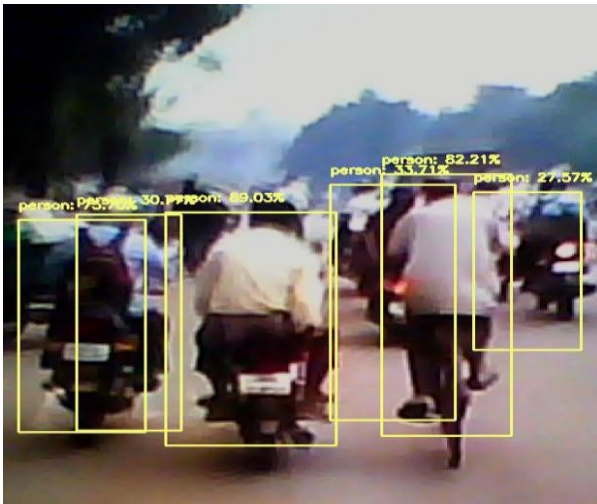


Fig. 8. Real Time validation of pedestrian detection

Even with a heavy background as captured in Fig. 8 and severely overlapping targets the model accurately predicts the target class.



Fig. 9. Detection of Pedestrians on a busy traffic signal

The results in Fig. 9 indicates that the model performs exceptionally well in heavy backgrounds and also takes into account the fact that real life scenarios have dynamically changing scenes and background and as such this model provides both an accurate and fast detection of pedestrian deployable in real time applications. The final result is a deep learning model efficient enough to process minimum 13FPS.

IV. CONCLUSION

The proposed model for pedestrian detection using MobileNet and Single Shot Detector is ideal for real time applications. Further improvements can be sought by the application of skip frames or using a variation of MobileNet to increase FPS throughput.

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